

Randy Pahang II

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EDUCATION

University of Nevada, Las Vegas

B.S. Computer Science — Honors College — Wolzinger Family Engineering Scholar

Expected May 2027

GPA: 4.0/4.0

EXPERIENCE

JT4 (Amentum)

May 2026 – Present

Systems Software Engineering Intern (Full-Time / Part-Time)

Las Vegas, NV

- Built a **C++23 telemetry simulator** generating **1,000+ packets/sec** in real time to stress-test downstream processing systems, replacing legacy commercial software and saving **\$10K+ in licensing costs**.
- Designed a configurable packet-generation pipeline reproducing **flight-representative timing and data rates** to validate downstream telemetry processing before flight-test and range operations.
- Exposed the C++ simulation engine through a **C ABI** to a Python/PySide6 dashboard for real-time inspection of **100+ telemetry channels**.
- Validated generated telemetry against recorded flight-test data and built a GoogleTest suite with **93% coverage** across packet serialization, timing, and protocol edge cases.

STARS Solution LLC

Nov 2025 – Jun 2026

Software Engineering Intern (Part-Time)

Las Vegas, NV

- Designed and built **Java Spring Boot** backend services integrating institution data from **30,000+ schools** for a **3,500+ user pilot**.
- Reduced AI recommendation time-to-first-response from 8.2s to 0.4s by decoupling generation with **RabbitMQ** and streaming partial results via **Server-Sent Events**.
- Reduced institution-search latency from 110ms to 37ms using a **Redis cache-aside** layer with versioned invalidation and database fallback.
- Improved recommendation **precision@5 by 22%** on a labeled evaluation set by filtering ineligible candidates and reranking retrieved results before generation.

LEADERSHIP

Bit2Byte

Nov 2025 – Present

Co-Founder & Lead Developer

Las Vegas, NV

- Co-founded UNLV's software development organization, growing to 100+ members and providing technical direction for student teams building with **React/Next.js, Spring Boot, and FastAPI**.
- Led technical workshops, sprint planning, code reviews, and project coordination across student development teams.

PROJECTS

PortRN | *Java Spring Boot, AWS ECS, RabbitMQ, Stripe*

Nov 2025 – Apr 2026

- Built and launched a professional networking platform with a **Spring Boot** backend deployed on **AWS ECS**, serving a 50+ user beta cohort.
- Parallelized multi-step AI portfolio generation across asynchronous **RabbitMQ workers**, reducing end-to-end processing from 7–10 to 4–5 minutes.
- Implemented **Stripe webhook** fulfillment with signature verification and **idempotency** to safely handle retries and prevent duplicate transactions.

HotelGuard | *1st Place, RebelHacks 2026*

Jan 2026

- Won **1st place out of 100+ participants** by building a real-time hotel risk-detection platform.
- Built a concurrent pipeline using **YOLOv8 tracking** and parallel LLMs to generate room-level risk scores.

TrafficControl | *Go, TCP/UDP, Linux, YAML*

Sep 2026 – Present

- Engineered a high-throughput network traffic simulator in **Go**, orchestrating concurrent **TCP/UDP** streams driven by declarative **YAML** configurations.
- Designed a concurrent engine utilizing **goroutines and channels** for stream lifecycles and graceful shutdowns, verifying thread safety with Go's race detector.

TECHNICAL SKILLS

Languages: C/C++, Java, Python, Go, TypeScript, SQL

Frameworks & Data: Spring Boot, FastAPI, Node.js, React, Next.js, PostgreSQL, Redis, RabbitMQ

Cloud & Systems: AWS (ECS, EC2, S3), Docker, Linux, Git, CI/CD, CMake, GoogleTest